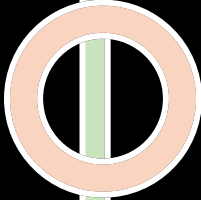


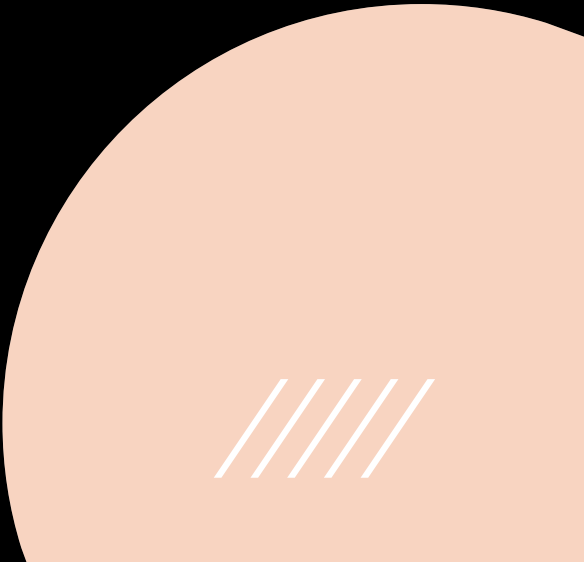
Fault Trees

Christina Kolb



Recap:

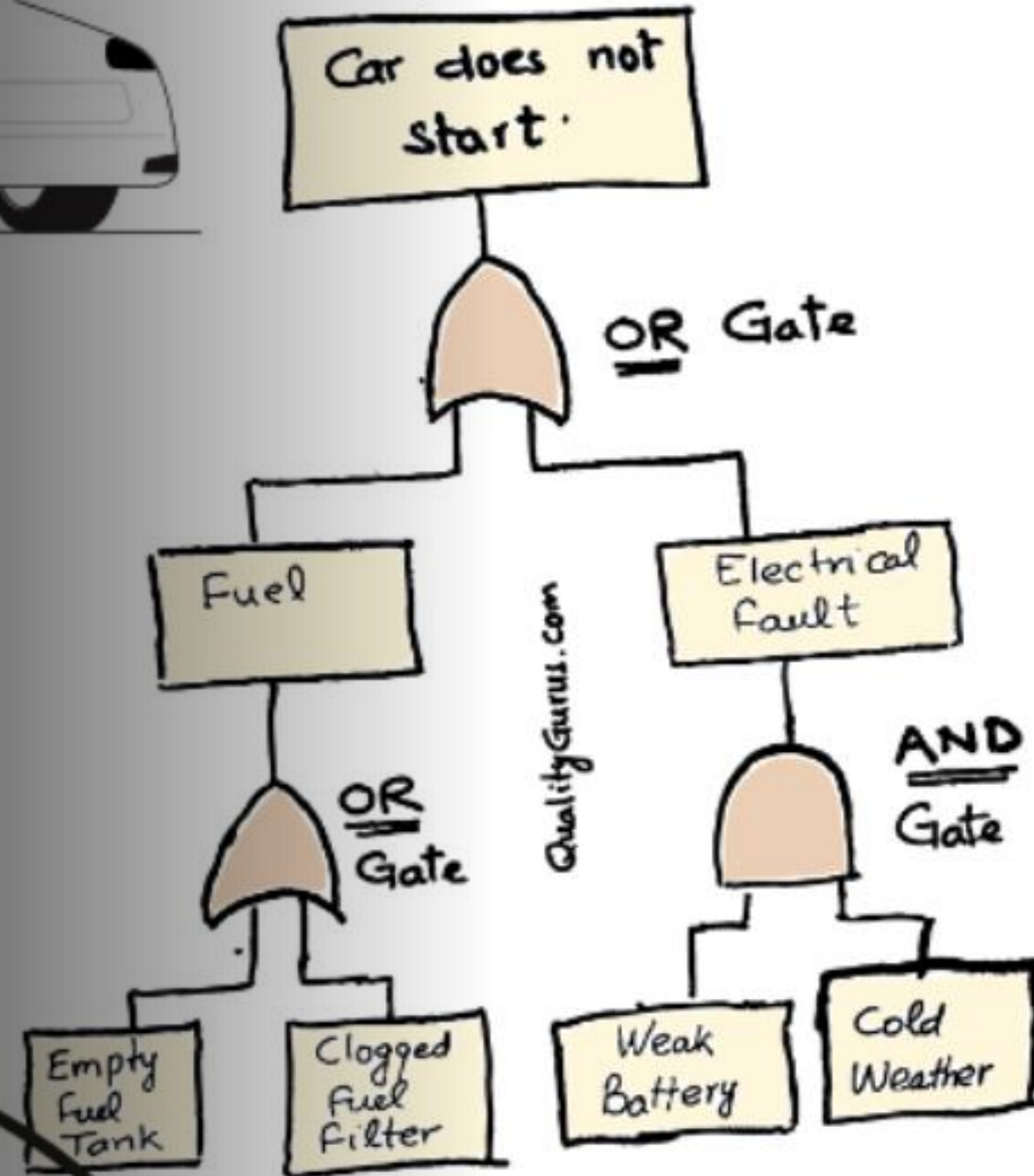
What do you
know about
Attack Trees?



Fault Tree Example



FAULT TREE ANALYSIS (FTA)



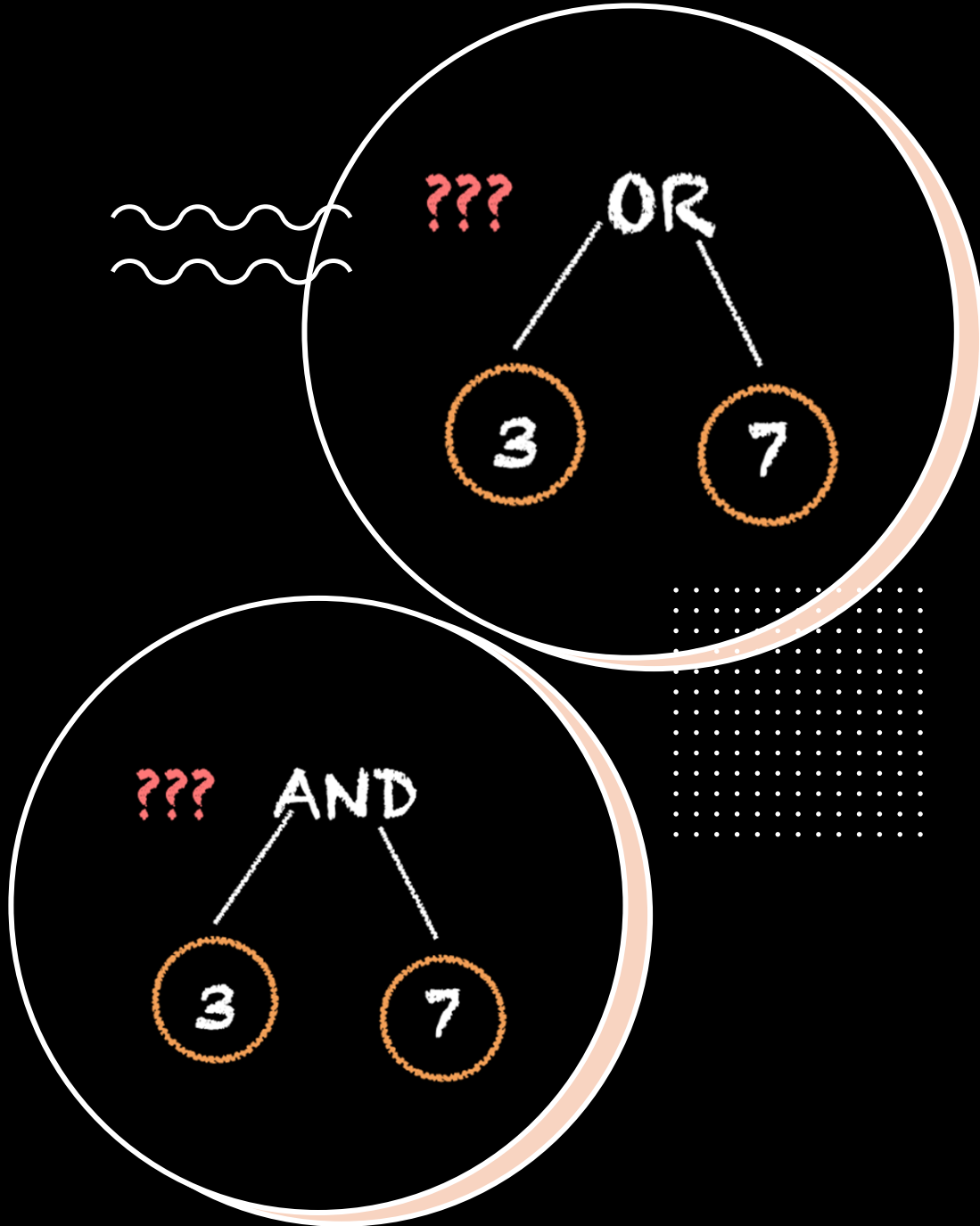
Fault Trees

Origin:

- Introduced in 1961 at Bell Labs to study ballistic missiles
- To evaluate the Minuteman I launch control system

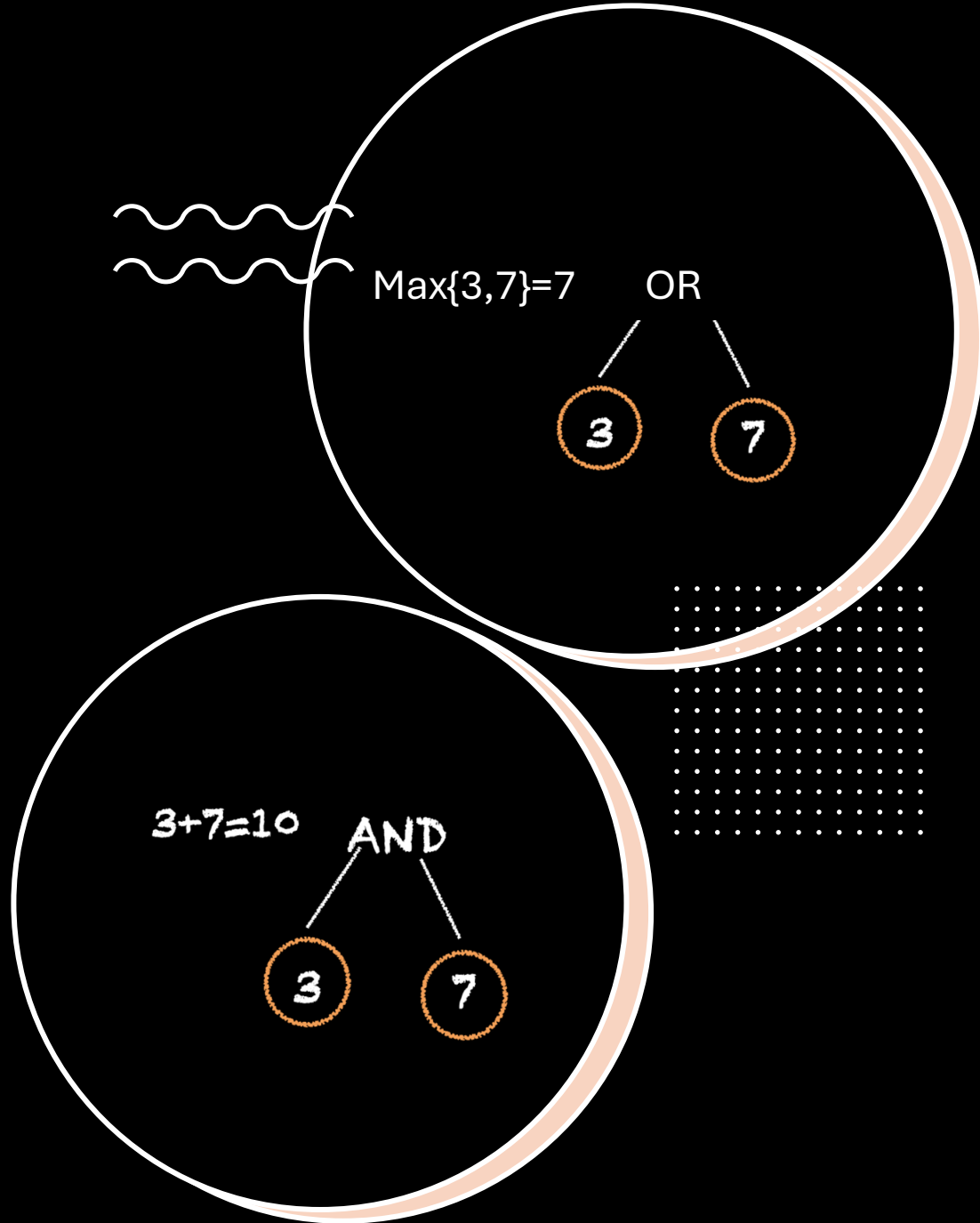
Structure:

- Basic events
- Intermediate Events
- Top Event
- And/ Or gates



Propagation of Values Through Logical Gates in Fault Trees

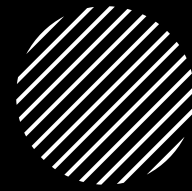
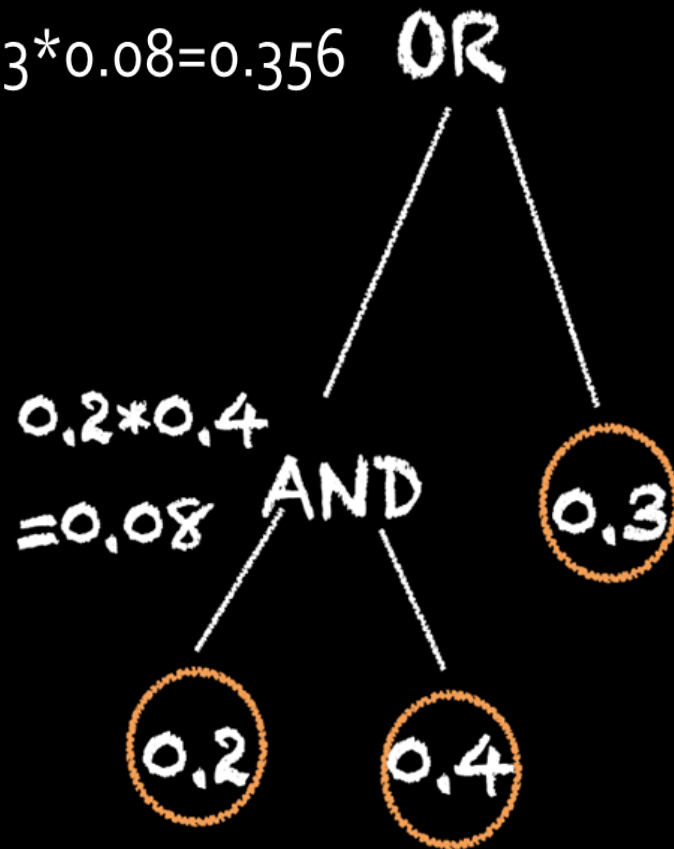
- Quantitative queries can be computed bottom-up.
- The values of the basic elements are propagated upwards in the tree
- Example:
Basic elements a and b cost resp. 3 EUR and 7 EUR



Propagation of Values Through Logical Gates in Fault Trees

- Quantitative queries can be computed bottom-up.
- The values of the basic elements are propagated upwards in the tree
- Example:
Basic elements *a* and *b* cost resp. 3 EUR and 7 EUR, then the cost of $\text{or}(a, b)$ is the max, 7 EUR, and the cost of $\text{and}(a, b)$ is the sum, 10 EUR.

$$0.08 + 0.3 - 0.3 * 0.08 = 0.356$$



Total Probability of Failure

- The value of an or gate in a fault tree is the sum of the values of its children, minus the value of their intersection.
- The values in the basic elements are given
- The probability of an and gate is the product of its children

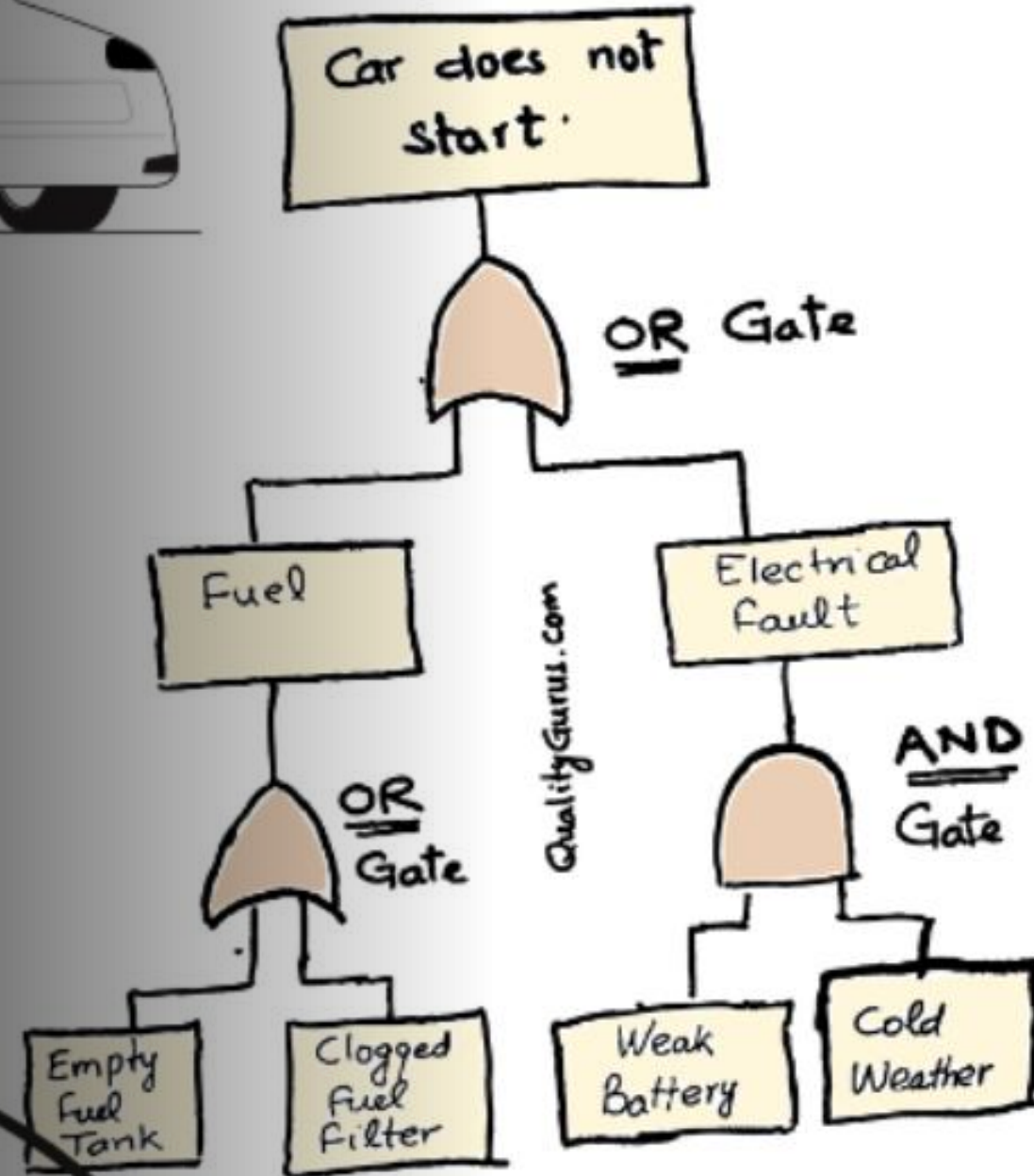
What are the similarities between attack trees and fault trees? - Show examples

What are the differences between attack trees and fault trees? - Show examples

Fault Tree Example



FAULT TREE ANALYSIS (FTA)



Tutorial



Model a case study of your choice with a fault tree.



Find and present some extensions of fault trees.

Report Information

- Content: (1) Scenario about security and safety problems, (2) Attack tree, (3) fault tree, (4) according to your case how does a at-ft combination looks like, (5) Improve your phishing mail to make it more convincing based on what you learned, (6) How would AI change malware detection capability and decision making, on both offensive and defensive sides?, (7) Economics and cybersecurity
- Format: title, name, intro, content, your findings/ conclusion, bibliography
- 6-8 pages
- AI statement: if you used AI for what and which tool
- Send an e-mail with attachment at c.kolb@utwente.nl
- Deadline end of september